

Form Name:	FSE UG and PGT Ethics Application
Submission Time:	April 23, 2026 12:41 pm
Browser:	Firefox 149.0 / Windows
IP Address:	49.37.32.163
Unique ID:	1454100913
Location:	

Faculty of Science and Engineering UG and PGT Ethics Application

To continue, you must confirm that you have read and understood the University of Hull Research and Governance Policies regarding Ethics.

I can confirm that I have read and understood the relevant policies

Applicant Information

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Project title	Sustainable Crop Yield prediction via Multi-modal AI
Module Number	6

Please write a brief summary of your research aim(s), objective(s), question(s) and/or hypothesis(es).

Background: Accurate crop yield forecasting is vital for food security and climate adaptation. Modern AI uses satellite and weather data but heavily relies on unimodal or naively fused models, making them fragile to cloud occlusion, missing weather readings, or sensor noise. Satellites suffer frequent cloud obstruction, while weather models lack the spatial detail needed for localized crop stress. In practice, agricultural data is often incomplete and noisy. This project addresses a fundamental AI limitation: the lack of robustness under modality degradation. Aim: To develop and rigorously evaluate a Gated Multimodal Deep Learning architecture that dynamically adapts to corrupted or missing input modalities for reliable crop yield prediction.

Objectives:

1. Primary: Create a multimodal mixture-of-experts model to evaluate unimodal baselines (satellite/weather), early/late-fusion baselines, and a Gated Multimodal Unit (GMU) that dynamically weights modalities. The model will undergo controlled modality dropout to simulate sensor failure, quantifying robustness across increasing degradation levels (10%, 25%, 50%, 75%) and comparing geographic generalization.
2. Secondary: Build a deployable Python library exposing the model and its configurations.

Novelty & Contribution: The innovation extends beyond standard multimodality to include explicit modeling of modality reliability, controlled robustness stress-testing, benchmarking operational resilience rather than ideal-case accuracy, and an architectural bias tailored for real-world agricultural deployment.

Research Questions:

1. What is the quantifiable performance of top unimodal models (3D-CNN for satellite vs. Bi-LSTM for weather), and under what specific conditions (e.g., >20% clouds, extreme drought) do they statistically fail?
 2. Does a gated multimodal architecture significantly reduce performance degradation under simulated dropout compared to naive fusion?
 3. Can an automated quality check sustain accurate predictions even if 50% of satellite imagery is cloud-covered?
 4. To what extent does a GMU reduce error variance (RMSE) compared to baselines under non-stationary noise, establishing a new operational benchmark?
 5. Is deep feature-level fusion superior to raw data-level fusion for spatiotemporal agricultural prediction?
 6. Can a dual-stream architecture with intermediate gating autonomously separate data quality from feature importance without explicit supervision, providing interpretability?
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Please add your project start date: Jan 2026

Please add your project end date: Sep 2026

Is ethical approval already in place for this research project? No

Research Methodology

Type(s) of research methods to be used Research using secondary data that includes no personal information about identifiable people
Research outside the UK

Details of Research Methods

Please outline your planned research methodology:

Rationale: This methodology mimics a human expert agronomist. An expert doesn't just look at a field (Satellite) or just read a thermometer (Weather), they synthesize both. If they can't see the field due to fog, they rely on their knowledge of the temperature. This project encodes that resilience into a mathematical model.

Data Selection and Collection: The project uses the CropNet Benchmark (2023), a standard public collection of agricultural data:-

- Input A (Visual): Satellite images (Sentinel-2) that show how green/healthy the plants are over time.
- Input B (Environmental): Daily weather records (Temperature, Rainfall, Sun).
- Target: Historical records of how much crop was actually harvested (Yield).

Analytical Process:

- 1) Alignment: Match the daily weather records with the weekly satellite images so they cover the exact same time periods.
- 2) Pattern Recognition: Use software to identify growth trends in the images and stress factors (like drought) in the weather data.
- 3) Smart Fusion: Implement the smart reliability filter to weigh the importance of each source. If it sees clouds, it trusts the weather more. If the weather data is missing, it trusts the images more.
- 4) Stress Testing: Intentionally delete parts of the data (simulating sensor failure) to see if the system can still make good predictions.

Experimental Design:

Baseline model to measure performance against:

1. CNN (Satellite-only)
2. LSTM (Weather-only)
3. Early Fusion (concatenated raw inputs)
4. Late Fusion (separate encoders + MLP)

Proposed model: 3D-CNN and LSTM/Transformer combined with a Gated Multimodal Unit (GMU) that dynamically adjusts modality importance per sample.

Robustness Testing

Simulated degradation scenarios:

Scenario Description

Image dropout Random cloud masking

Weather dropout Random feature masking

Structured dropout Seasonal data removal

Combined dropout Multi-modality corruption

Performance degradation curves will be analyzed.

Evaluation Metrics

- RMSE
- MAE
- R^2
- MAPE
- Relative Degradation Index (RDI)

- Robustness Retention Score (RRS)

Statistical validation:

- Paired t-tests
- 5-fold cross-validation
- Geographic holdout validation

Please describe the ethical issues arising from your research plans and how they will be addressed:

The proposed system predicts county-level soybean yields (bushels per acre) across the contiguous United States from 2016 to 2022 by fusing USDA NASS census data, high-resolution WRF-HRRR atmospheric reanalysis, and Sentinel-2 optical satellite imagery.

Because agricultural predictions directly influence global food security policies and climate adaptation strategies, this project falls under a Ethics Risk Tier according to the EU AI Act (2024).

The system must be audited against stringent governance frameworks, including the NIST AI Risk Management Framework (RMF) 1.0, IEEE P7003 (Algorithmic Bias Considerations), and the ACM FAccT Principles.

In multimodal AI for agriculture, biases typically manifest through structural disparities in data collection and environmental phenomena.

- **Geographic Representation & Infrastructure Bias:** The data is skewed within certain regions will lead to bias. However the data successfully maps 15 Sentinel FIPS codes to 105 matching USDA records.

- **Sensor Occlusion & Measurement Bias:** Real-world agricultural data is notoriously noisy, and optical satellite systems are frequently obstructed by cloud cover. Single sensors struggle with the multidimensional complexity of agricultural fields due to their inherent incompleteness. Furthermore, optical satellite technologies have a pronounced sensitivity to variable weather conditions, which can substantially influence data reliability. Missing sensor inputs and poor data quality remain significant adoption barriers and ethical concerns for AI in precision agriculture.

- **Confounding Temporal Anomalies:** Including all 7 years of data maximizes the training signal but more data could be required.

3. Mathematical Resolution and Quantitative Fairness Strategies

To counter these systemic biases, the project implements rigorous architectural and programmatic interventions.

- **Architectural Fairness (The GMU):** To address sensor bias, the architecture employs a Gated Multimodal Unit (GMU). This mechanism dynamically shifts attention weights to rely more heavily on unimodal weather data if satellite inputs are masked by clouds. Integrating heterogeneous data sources through multimodal fusion enhances the comprehensiveness and robustness of the system. Furthermore, combining sensor inputs into hybrid multimodal datasets has been proven to enhance the accuracy of statistical yield-prediction models.

- **Programmatic Fairness Metrics:** The pipeline is designed specifically to measure equity across different regions, including sliced_rmse, bias_per_group, cv_error_per_state, and a holistic fairness_report.

- **Target Transformation and Jensen Correction:** The target variables are log-transformed to benefit computational

optimization. To preserve ethical interpretability and report accurate yields, the framework mandates the use of a mathematically rigorous Jensen correction constant ($JENSEN_SIGMA2$).

Though the project deals with a important global issue regarding agriculture yield and any biases can affect the outcome but if necessary steps are taken to mitiage bias and ensure fairness, then these project would be risk free and ethically important

Research outside the United Kingdom

Where do you plan to conduct your research?

India

Please indicate whether you or your supervisor have checked if there are ethical review procedures in the country, or countries, involved that you will need to comply with. If so explain how you will comply with these procedures

Yes, my supervisor and I have reviewed the ethical and legal procedures relevant to the multiple jurisdictions involved in this research. Because this project is computationally executed in India, governed by a UK institution, and utilizes data from the United States, we have assessed the compliance requirements for all three regions:

1. Data Origin Country (United States):

The research relies on the CropNet dataset, which synthesizes US-based data including USDA NASS census records, NOAA HRRR weather data, and ESA Sentinel-2 imagery. Because this is open-source, publicly available macroeconomic and meteorological data that does not involve human subjects, animal testing, or Personally Identifiable Information (PII), it does not trigger the need for a US Institutional Review Board (IRB) approval. My compliance in this jurisdiction is strictly limited to adhering to the open-source data licenses and terms of use stipulated by the USDA, NOAA, and ESA.

2. Institutional Country (United Kingdom):

As a researcher affiliated with the University of Hull, the primary ethical authority governing this project is the UK. I am complying with the University's Ethics Code of Practice and will not proceed to full data publication or deployment until this formal ethics review is approved by the University's ethics committee.

3. Country of Execution (India):

I am physically conducting the data processing and model training from India. Because the dataset contains US agricultural data and zero personal data, it does not fall under the purview of India's Digital Personal Data Protection (DPDP) Act. However, I will maintain strict data security and research integrity protocols (such as securing API credentials and removing hardcoded keys) on my local machines to comply with standard academic cybersecurity practices.

Please indicate whether you or your supervisor have checked if there are local laws or regulations that may impact on your research, and if so explain how you will comply with this legislation

Yes, my supervisor and I have reviewed the local laws and regulations across the applicable jurisdictions. Because the computational execution and model training for this research take place locally in West Bengal, India, while the project is governed by a UK academic institution and utilizes international data, we have structured our compliance as follows:

1. Local Indian Legislation:

The primary local regulations concerning computational research and data handling are the Information Technology Act, 2000, and the Digital Personal Data Protection Act (DPDPA), 2023. Because this research exclusively utilizes macro-agricultural census records (USDA NASS), atmospheric reanalysis (WRF-HRRR), and satellite imagery (Sentinel-2), it processes absolutely no Personally Identifiable Information (PII), human participant data, or sensitive local data. Therefore, it is exempt from the restrictive data-processing clauses of the DPDPA. To comply with general local cybersecurity mandates regarding research integrity, we are actively implementing security protocols to ensure all API keys (such as the Sentinel API credentials) are removed from local Jupyter notebooks and secured via environment variables to prevent unauthorized access or local data breaches.

2. UK Institutional Governance:

As this dissertation is conducted under the University of Hull, it is subject to the UK Data Protection Act 2018 and UK GDPR. Compliance is straightforward as no personal data is generated or stored. We will fully comply with the University's Research Data Management policies by ensuring the final Gated Multimodal Unit (GMU) architecture and aligned datasets are preserved and shared openly according to standard academic open-source licensing.

3. International AI & Algorithmic Legislation:

Because this system predicts county-level agricultural yields, its outputs have the potential to influence macroeconomic agricultural planning and food security. As a result, it intersects with emerging international algorithmic laws, specifically the EU Artificial Intelligence Act (Regulation (EU) 2024/1689), which classifies such predictive systems as a High Ethics Risk Tier. To comply with these strict regulatory standards for high-risk AI, the project mandates a formal fairness contract. We are complying by technically embedding a bias registry (B1-B9) and automated programmatic fairness metrics (such as `sliced_rmse` and `bias_per_group`) directly into the validation pipeline to mathematically prove the model does not discriminate against specific geographic regions.

If you will be working with local organisations and/or individuals, please describe any ethical considerations and how these will be addressed

I will not be working with any local organizations, industry partners, or individual participants during the course of this research.

This project is entirely computationally based and relies exclusively on the secondary analysis of open-source, publicly available data, specifically the CropNet benchmark, USDA NASS agricultural census records, NOAA HRRR atmospheric reanalysis, and ESA Sentinel-2 satellite imagery.

Because there is no primary data collection involving human subjects, no interviews or surveys with local farmers, and no physical deployment of the system through local agricultural organizations, traditional ethical considerations relating to informed consent, participant confidentiality, physical harm, or organizational exploitation are not applicable to this study. The ethical scope of this research is strictly confined to algorithmic accountability, algorithmic fairness, and mitigating geographic or sensor biases within the deep learning architecture.

Data Management Plan

When you conduct your research, will you be generating new information or data? No

Submitting your Ethics Application
